THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 05

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 13 units	$\sqrt{25+144}=13$
2	**(a)** $6n-5$	common difference 6, so $a_n=6n-5$
3	**(d)** $\frac{1}{7}$	$\frac{1}{7}=0.\overline{142857}$
4	**(d)** $x^2+15x+56$	$(x+a)(x+b)=x^2+(a+b)x+ab$
5	**(a)** 12 cm	half-chord $=\sqrt{10^2-8^2}=6$, so chord $=2\times6=12$
6	**(b)** 6 cm^2	$s=6$, area $=\sqrt{6\cdot3\cdot2\cdot1}=6$
7	**(a)** 0.23	$46/200=0.23$
8	**(c)** $\frac{3}{5}$	9 favourable out of 15 equally likely balls
9	**(b)** 39	$a+(n-1)d=-5+11\times(4)=39$
10	**(b)** 12	it satisfies the ‘if’ part but not the ‘then’ part
11	**(b)** may be true or false	e.g. “if it rains, the road is wet”
12	**(b)** 45%	cooling share $=\frac{450}{1000}\times100=45\%$
13	**(c)** 10%	$\dfrac{600\times8+400\times13}{600+400}=10\%$
14	**(b)** 335	the leading 1 of 1335 is lost
15	**(b)** all its internal angles are less than 180°	a ‘dent’ makes an angle exceed 180°
16	**(b)** 21 cm	AG = 2·GD, so AD = AG + GD = 3·GD
17	**(a)** 74	x + y = 11, x − y = 3 give x = 7, y = 4
18	**(d)** A is false but R is true	A is false (slope is undefined); R is true
19	**(d)** 440 cm^2	$2\pi rh=440$
20	**(a)** Both A and R are true and R is the correct explanation of A	e.g. $8\times8\times8$ and $16\times8\times4$ both have volume 512 but TSA 384 and 448

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$4-15=-11$ °C.	1+1
22	$d=\sqrt{13^2-5^2}=12$ cm for each; equal chords are equidistant from the centre.	1+1
23	Arc $=\frac{1}{4}\times2\times\frac{22}{7}\times14=22$ cm; perimeter $=22+28=50$ cm.	1+1
24	$1-0.35=0.65$, since the win and not-win events together are certain (probability 1).	1+1
25	For $n=4$: $2^4+1=17$ is prime, but for $n=6$: $2^6+1=65=5\times13$ is composite (2).	1+1
26	$360^\circ-(85^\circ+95^\circ+110^\circ)=360^\circ-290^\circ=70^\circ$ (2).	1+1
27	Slope $=\dfrac{3}{12}=\dfrac{1}{4}$ (1). For every 4 m moved horizontally the ramp rises 1 m (1).	1+1
28	$V_{cone}:V_{cyl}=\frac{1}{3}\pi r^2h:\pi r^2h$ (1) $=1:3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Pairs: $(1,36),(2,18),(3,12),(4,9),(6,6)$ (2). Only numbers up to $6=\sqrt{36}$ need to be examined (1).	2+1
30	$500-20d$; zero when $d=25$ days. After 15 days, 200 pages remain.	2+1
31	(i) $3^2:5^2=9:25$ (1.5). (ii) $3^3:5^3=27:125$ (1.5).	1½+1½
32	$15\div2=\frac{15}{2}=7\frac{1}{2}$: 7 kurtas, with 1 m left over.	1+1+1
33	Half-perimeter 18; the maximum area occurs for the square: side 9 cm (2); area 81 cm^2 (1).	2+1

34	(a) Yes, the sticks must be equal since the diagonals are the sticks (1½). (b) Yes, it matters: if the sticks cross at their midpoints a rectangle results; if they cross elsewhere an isosceles trapezium can result, so equal sticks alone do not fix the shape (1½).	1½+1½
35	$P = \left(\frac{2(2)+8}{3}, \frac{2(-2)+4}{3}\right) = (4, 0)$. $AP = \sqrt{4+4} = 2\sqrt{2}$; $AB = \sqrt{36+36} = 6\sqrt{2}$; $AP = \frac{1}{3}AB$ ✓	2+1
	OR: $AB = \sqrt{9+16} = 5$, $AC = \sqrt{16+9} = 5$, $BC = \sqrt{49+1} = \sqrt{50}$. $AB = AC$ (isosceles) and $AB^2 + AC^2 = 50 = BC^2$ (right angle at A).	1+1+1
36	(i) 5, 10, 15, 20, 25 ⇒ $\frac{5}{25} = \frac{1}{5}$ (ii) 2,3,5,7,11,13,17,19,23 ⇒ $\frac{9}{25}$.	1+2
	OR: April, June, September, November ⇒ $\frac{4}{12} = \frac{1}{3}$ (3).	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Slope $= \frac{6-(-2)}{4-0} = 2$ (1). (ii) $y = 2x - 2$ (1). (iii) Put $y = 0$: $x = 1$, so $(1, 0)$ (1). (iv) $x + 2x - 2 = 4$, so $3x = 6$, $x = 2$ (1). $y = 2$, so they meet at $(2, 2)$; check $2 + 2 = 4$ (1).	1+1+1+1+1
	OR: (i) Slope $= 4$: the level rises 4 cm every minute (1). (ii) $y = 4(6) = 24$ cm (1). (iii) $4x = 50$, $x = 12.5$ minutes (1). (iv) $4(8) = 32 \neq 30$, so $(8, 30)$ is not on the line (1). (v) 10 is the y-intercept: the water level is already 10 cm at the start (1).	1+1+1+1+1
38	(i) 2 : 1, longer part towards the vertex (1). (ii) $AG = 18$, $GD = 9$, $BG = 24$, $GE = 12$ m (2). (iii) $CG = \frac{2}{3} \times 30 = 20$ m and $GF = 10$ m (1). (iv) Using two medians, G divides both in 2 : 1; the third median with either gives the same point because only one point divides a segment in a given ratio (1).	1+2+1+1
	OR: (i) Equal diagonals ⇒ PQRS is a rhombus; perpendicular diagonals ⇒ it is also a rectangle; so it is a square (2). (ii) Each side $= 35$ m (1). (iii) Perimeter $= 140$ m; area $= 35^2 = 1225$ m^2 (1). (iv) Field area $= \frac{1}{2} \times 70 \times 70 = 2450$ m^2 (1).	2+1+1+1
39	(i) $\lvert -4 - 3\rvert = 7$; $\left\lvert -\frac{5}{2} - \frac{7}{2}\right\rvert = 6$. (ii) $\frac{7}{9}$, 0, $\frac{5}{3}$. (iii) $\frac{13}{4} = 3.25$ lies between 3 and 4.	2+1+2
	OR: (i) $\frac{x}{2} = \frac{9}{2} \Rightarrow x = 9$. (ii) LHS $= \frac{1}{2} \times \frac{13}{12} = \frac{13}{24}$; RHS $= \frac{3}{8} + \frac{1}{6} = \frac{9}{24} + \frac{4}{24} = \frac{13}{24}$ ✓	2+3
40	(i) $200l + 160w + 50lw$. (ii) $4000 + 1600 + 10000 = ₹15,600$. (iii) Terms: $200l$, $160w$, $50lw$; variables l, w; coefficients 200, 160, 50.	2+2+1
	OR: (i) $x + 4x = 300 \Rightarrow$ 60 ft and 240 ft. (ii) $2(3w + 3) = 24 \Rightarrow w = 3$ m, $l = 9$ m. (iii) 27 m^2.	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.